Day-14 (Nesting of Member function)

Nesting of member function means calling one member function of same class.
Simply class has many functions in which one function can call another function.

Why nested is used

Nesting is used to to make program organized and secure.
Main Reason:

1. Code Organization.
   - Large tasks are devided into small functions.
2. Reusability
   - One function can be reused multiple times.
3. Data Protection
   - Private function can only be used inside the class.

Basic structure

```
Class Classname {
    data members;
        func1();
        func2();
        func3();
}
```

inside a function

```
function1() {
    func2(); // nested call.
}
```

Simple Example

```
#include <iostream>
using namespace std;
class Number {
    int n;
```

Public :

```
void show() {
        secret(); // allowed (nesting)
    };
};

int main() {
    Example obj;
    obj.show();  //works
}
```

but
obj.secret(); X

Problem Statement: Binary Checker Program (C++)

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Binary these are not

contains only 1 and 0;

```
Public :
    void getNumber () {
        cout << "Enter a number ;";
            cin >>n;
    }
    void CheckEven () {
        if (n % 2 == 0) {
            cout << " Number is Even ";
            Else
              cout << " number is odd " ;
            }
        void result () {
            CheckEven () ;   // nested member function.
        }

};
int main () {
    Number obj ;
    obj. getNumber() ;
    obj. result ();
    return 0 ;
}
```

How it will proceed

Program start from main function

Object is created (Number obj;)

First function runs : obj. getNumber () ; user enter number

Second function runs obj. result () ; inside result CheckEven () ; is called

Important rule

A private function can also be used in nesting

But it cannot be called outside the class.

Ex
```
#include <iostream>
using namespace std ;
class Example {
    private :
        void secret () {
            cout << " This is a private function " ;
        }
```